An Analytic Hazard Kernel for Optogenetically-Driven Larval Reorientation

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Abstract

Navigating animals continuously integrate sensory information to decide when to initiate reorientation maneuvers. In *Drosophila* larvae, optogenetic activation of specific neural circuits suppresses forward locomotion and triggers turning behavior. We present an analytic hazard model for predicting the timing of reorientation events under controlled LED stimulation. The model parameterizes the temporal response as a difference of two gamma probability density functions, capturing a fast sensory transduction component ($\tau_1 = 0.29$ s) and a slower synaptic adaptation component ($\tau_2 = 3.81$ s). Fit to 1,407 events from 55 larval tracks, the 6-parameter kernel achieves $R^2 = 0.968$ against a 12-basis raised-cosine reference and reproduces empirical event rates with a ratio of 0.97. We demonstrate the model's application in a RUN/TURN trajectory simulator that matches observed turn rates (1.88 vs 1.84 turns/min). This interpretable formulation enables quantitative comparison across experimental conditions and provides a foundation for biologically-grounded navigation simulations.

1 Introduction

1.1 Larval Navigation and Optogenetic Control

Drosophila larvae navigate their environment using a characteristic locomotor pattern of forward crawling (`runs'') punctuated by reorientation maneuvers (`turns'') during which the animal samples new heading directions (Gomez-Marin et al., 2011). These turns are not random; their timing and direction are modulated by sensory input, enabling larvae to perform gradient climbing, odor tracking, and phototaxis (Gershow et al., 2012; Kane et al., 2013).

Optogenetic tools provide precise temporal control over neural activity, allowing researchers to probe how specific circuits influence behavioral decisions. In GMR61 larvae expressing channelrhodopsins, LED illumination activates neurons that suppress forward locomotion and increase the probability of reorientation events (Gepner et al., 2015). Understanding the temporal dynamics of this suppression—how the probability of initiating a turn evolves after stimulus onset and offset—is central to modeling sensorimotor integration.

1.2 The Need for Interpretable Hazard Models

Previous work has modeled larval turning probability using linear filter models and generalized linear models (GLMs) with temporal basis functions (Hernandez-Nunez et al., 2015; Klein et al., 2015). These approaches fit flexible kernels—often using raised-cosine or spline bases with many parameters—that capture the temporal structure of stimulus-response relationships.

While flexible basis representations achieve good predictive performance, they offer limited interpretability. A 12-parameter raised-cosine kernel may fit the data well but does not directly reveal

the underlying timescales, nor does it distinguish contributions from distinct biological processes (e.g., sensory transduction vs. adaptation).

1.3 Contribution

We address this gap by developing an **analytic hazard kernel** for optogenetically-driven reorientation. Our kernel is a difference of two gamma probability density functions:

$$K_{\text{on}}(t) = A \cdot \Gamma(t; \alpha_1, \beta_1) - B \cdot \Gamma(t; \alpha_2, \beta_2) \tag{1}$$

where $\Gamma(t; \alpha, \beta)$ denotes the gamma probability density function with shape α and scale β.

This 6-parameter form captures:

1. A **fast excitatory component** (peak ∼0.16 s, mean $\tau_1 = 0.29$ s) representing rapid sensory transduction
2. A **slow suppressive component** (peak ∼2.9 s, mean $\tau_2 = 3.81$ s) representing synaptic or network adaptation

The gamma-difference form is not arbitrary: it arises naturally as the impulse response of cascaded first-order processes, making parameters directly interpretable in terms of processing stages and time constants.

We validate this kernel against experimental data from GMR61 larvae under 10 s ON / 20 s OFF LED stimulation, demonstrating that:

- The analytic kernel approximates a 12-basis reference with $R^2 = 0.968$
- The hazard model reproduces empirical event rates (rate ratio = 0.97)
- A RUN/TURN trajectory simulator driven by the model matches observed behavior

2 Methods

2.1 Experimental Data

Data were collected from GMR61 *Drosophila* larvae expressing channelrhodopsins. Animals were tracked at 20 Hz on an agar substrate while receiving optogenetic LED stimulation in a square-wave pattern: 10 s ON at 250 PWM, 20 s OFF (30 s cycle). For this study, we analyzed **55 tracks** containing **1,407 reorientation-onset events** from 2 experimental sessions under the 0 → 250 PWM, control (7 PWM background) condition.

Reorientation events were detected using a curvature-threshold algorithm that identifies the onset of heading changes. This inclusive definition captures both large sustained turns and brief head sweeps; events with measurable duration >0.1 s were classified as "true turns'' ($N = 319$) for behavioral interpretation.

2.2 Hazard Model Structure

We model reorientation timing as a point process with instantaneous hazard:

$$\lambda(t) = \exp\left(\beta_0 + u_{\text{track}} + K_{\text{on}}(t_{\text{onset}}) + K_{\text{off}}(t_{\text{offset}})\right) \tag{2}$$

where:

- $\beta_0 = -6.23$: Calibrated global intercept (log-hazard baseline)
- $u_{\text{track}} \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 0.47$: Track-specific random effect capturing individual variability
- $K_{\text{on}}(t)$: LED-ON kernel response to stimulus onset
- $K_{\text{off}}(t)$: LED-OFF kernel response to stimulus offset

The model is fit as a negative-binomial GLM (NB-GLM) with logarithmic link, treating each video frame ($dt = 0.05$ s) as a Bernoulli trial for event occurrence.

2.3 LED-ON Kernel: Gamma-Difference

The LED-ON kernel is parameterized as a difference of two gamma probability density functions (Equation~1), where:

$$\Gamma(t; \alpha, \beta) = \frac{t^{\alpha-1} e^{-t/\beta}}{\beta^\alpha \Gamma(\alpha)} \tag{3}$$

is the gamma PDF with shape α and scale β.

Table 1: Fitted kernel parameters with 95% bootstrap confidence intervals.

Parameter	**Value**	**95% CI**	**Interpretation**
A	0.456	[0.409, 0.499]	Fast component amplitude
α_1	2.22	[1.93, 2.65]	Fast shape (∼2 stages)
β_1	0.132 s	[0.102, 0.168]	Fast timescale
B	12.54	[12.43, 12.66]	Slow component amplitude
α_2	4.38	[4.30, 4.46]	Slow shape (∼4 stages)
β_2	0.869 s	[0.852, 0.890]	Slow timescale

The derived timescales are:

- Fast component: peak at 0.16 s, mean $\tau_1 = \alpha_1\beta_1 = 0.29$ s
- Slow component: peak at 2.94 s, mean $\tau_2 = \alpha_2\beta_2 = 3.81$ s

2.4 LED-OFF Rebound Kernel

A separate exponential kernel captures transient effects after LED offset:

$$K_{\text{off}}(t) = D \cdot \exp(-t/\tau_{\text{off}}) \tag{4}$$

with $D = -0.114$ and $\tau_{\text{off}} = 2.0$ s. This modest negative term represents continued suppression during recovery, with a half-life of 1.39 s.

2.5 Event Definition

The hazard model was fit to **all 1,407 inclusive onset events**, which include:

- Large, sustained reorientations ("true turns")
- Brief head sweeps and micro-movements
- Frame-by-frame curvature fluctuations

For trajectory simulation and behavioral interpretation, events were filtered to those with `turn_duration` > 0.1 s ($N = 319$, 23% of total). This two-stage approach follows standard practice in larval navigation modeling: hazard fitting uses the full temporal structure while behavioral output focuses on salient events.

For the factorial extension (Section~3.4), we pooled 12 experiments comprising 7,288 events across 623 tracks. Two experiments with anomalously high event counts (approximately 10–20× other sessions) were excluded because their annotation statistics were inconsistent with the remaining dataset.

2.6 Rate Calibration

The NB-GLM intercept ($\beta_0 = -6.76$) represents log-hazard per frame at 20 Hz. Discrete-time simulation with this intercept produced $ $60% of empirical events. We applied a calibration factor:

$$\beta_0^{\text{cal}} = \beta_0 + \log\left(\frac{N_{\text{emp}}}{N_{\text{sim}}}\right) = -6.76 + \log(1.695) = -6.23 \tag{5}$$

This global rate normalization preserves kernel shape, suppression timing, and relative condition effects while matching empirical event rates. All factorial contrasts are independent of this calibration.

2.7 Trajectory Simulation

We implemented a RUN/TURN state machine driven by the hazard model:

RUN state:

- Forward motion at 1.0 mm/s
- Brownian heading noise ($\sigma = 0.03$ rad/$\sqrt{\text{s}}$)
- Transition to TURN governed by hazard

TURN state:

- Angle sampled from $\mathcal{N}(\mu = 7^\circ, \sigma = 86^\circ)$
- Duration sampled from Lognormal (median = 1.1 s)
- Speed reduced to 0.4× run speed
- Return to RUN after duration elapsed

The trajectory simulator uses the hazard model to drive RUN/TURN transitions and reproduces event rates and timing. Spatial statistics of trajectories (path shapes, arena occupancy) were not systematically validated; the simulator is presented as a demonstration of possible use rather than a fully calibrated locomotion model.

2.8 Validation Metrics

We assessed model performance using:

1. **Kernel** R^2: Correlation between analytic and 12-basis raised-cosine kernels
2. **Rate ratio**: Simulated / empirical total events
3. **PSTH correlation**: Match between simulated and empirical peri-stimulus histograms
4. **Suppression magnitude**: Fold-change in event rate during LED-ON vs LED-OFF

3 Results

3.1 Analytic Kernel Captures Temporal Structure

The 6-parameter gamma-difference kernel closely approximates the 12-parameter raised-cosine reference (Figure~1A). The analytic form achieves $R^2 = 0.968$ and cross-validated $R^2 = 0.961$ (5-fold, track-wise), demonstrating that the compact parameterization does not sacrifice predictive accuracy.

The kernel shows characteristic biphasic dynamics visible in Figure~1B: an initial brief increase in hazard (fast component, $\tau_1 = 0.29$ s, red curve) followed by sustained suppression (slow component, $\tau_2 = 3.81$ s, blue curve). The fast component peaks at approximately 0.16 s post-stimulus, consistent with the latency of channelrhodopsin activation and first-order neural responses. The slow component dominates from 1–8 s, producing the characteristic suppression of reorientation probability during LED-ON periods. This biphasic pattern is consistent with rapid sensory transduction followed by slower synaptic adaptation.

The LED-OFF rebound kernel (Figure~1C) shows modest continued suppression after stimulus offset, with a time constant of 2.0 s. This indicates that the return to baseline behavior is gradual rather than instantaneous, likely reflecting recovery from synaptic adaptation.

3.2 Hazard Model Reproduces Event Rates

Simulation using the calibrated hazard model produces event counts closely matching empirical observations (Table~2; Figure~2).

Table 2: Validation metrics comparing simulated and empirical data.

Metric	**Empirical**	**Simulated**	**Status**
Total events	1,407	1,371	PASS
Rate ratio	—	0.974	Target: 0.8–1.25
LED-OFF rate	∼1.9/min	∼1.9/min	MATCH
LED-ON rate	∼1.0/min	∼1.0/min	MATCH
Suppression	2.0×	1.9×	MATCH

The peri-stimulus time histogram (PSTH) comparison reveals close agreement between model predictions and empirical observations (Figure~2A–B). The correlation between simulated and empirical event histograms is $r = 0.84$, indicating good capture of temporal dynamics around stimulus transitions. Notably, the model correctly reproduces:

- The rapid onset of suppression within 0.5 s of LED activation (Figure 2A, gray shaded region)
- The sustained $\sim 2\times$ reduction in event rate during the 10 s LED-ON period
- The gradual recovery to baseline following LED offset

The hazard function comparison (Figure~2C) shows strong correlation ($R^2 = 0.962$) between the analytic model's instantaneous hazard and the empirical event rate, confirming that the gamma-difference kernel captures the underlying temporal dynamics rather than merely matching aggregate statistics.

3.3 Trajectory Simulation Matches Behavioral Statistics

The RUN/TURN simulator driven by the hazard model produces realistic larval trajectories that recapitulate key features of empirical behavior (Figure~3; Figure~4). Key behavioral statistics match empirical observations:

Table 3: Trajectory simulation statistics.

Metric	**Simulated**	**Empirical**	**Match**
Turn rate	1.88/min	1.84/min	98%
Mean turn angle	7°	7°	MATCH
Turn duration	1.1 s median	1.1 s median	MATCH

Figure~3A shows example simulated trajectories over a 5-minute period, illustrating the characteristic pattern of extended RUN segments punctuated by discrete TURN events (marked by red circles). The spatial distribution of turns is modulated by LED state: during LED-ON periods (indicated by yellow background shading), turns are suppressed, leading to longer uninterrupted runs.

The turn angle distribution (Figure~4A) shows high variability ($\sigma = 86°$) with a slight rightward bias ($\mu = 7°$), consistent with the exploratory nature of larval navigation. The turn duration distribution (Figure~4B) follows a lognormal form with median 1.1 s, matching empirically observed turn durations. These distributions were extracted from the 319 filtered events (duration > 0.1 s) and used to parameterize the trajectory simulator.

3.4 Factorial Analysis of Intensity and Background Effects

To assess generalization beyond the reference condition, we extended the model to a 2×2 factorial design varying LED1 intensity ($0 \rightarrow 250$ vs $50 \rightarrow 250$ PWM) and LED2 background pattern (Constant 7 PWM vs Cycling 5–15 PWM). This analysis pooled 12 experiments comprising 623 tracks and 7,288 events across all four conditions (Figure~5).

The factorial model extends the hazard function with condition-specific modulation:

$$\log \lambda(t) = \beta_0 + \beta_I \cdot I + \beta_C \cdot C + \beta_{IC} \cdot (I \times C) + (\alpha + \alpha_I \cdot I + \alpha_C \cdot C) \cdot K_{\text{on}}(t) + \gamma \cdot K_{\text{off}}(t) \quad (6)$$

where I indicates the $50 \to 250$ intensity condition and C indicates the cycling background condition.

Table 4: Factorial model coefficient estimates with 95% confidence intervals. All effects are statistically significant at $p < 0.05$.

Effect	**Coefficient**	**95% CI**	**Hazard Ratio**	**Interpretation**
β_I (Intensity)	-0.199	$[-0.266, -0.132]$	$0.82\times$	Lower baseline hazard
β_C (Cycling)	-0.108	$[-0.174, -0.042]$	$0.90\times$	Lower baseline hazard
β_{IC} (Interaction)	-0.119	$[-0.218, -0.019]$	—	Synergistic reduction
α (Kernel amplitude)	1.005	$[0.899, 1.110]$	—	Baseline suppression
α_I (Intensity mod.)	-0.665	$[-0.773, -0.557]$	—	-66% suppression
α_C (Cycling mod.)	0.152	$[0.050, 0.254]$	—	$+15\%$ suppression
γ (Rebound)	1.669	$[0.470, 2.869]$	—	Post-offset enhancement

The factorial analysis reveals that both experimental manipulations significantly modulate the optogenetic response (Table~4; Figure~5):

Intensity effect ($50 \to 250$ vs $0 \to 250$): The $50 \to 250$ condition shows 66% weaker suppression amplitude ($\alpha_I = -0.665$, $p < 0.001$). This is consistent with partial adaptation: larvae pre-exposed to 50 PWM baseline illumination exhibit reduced sensitivity to the subsequent intensity step, suggesting that the sensory pathway has already partially adapted to the ambient light level.

Cycling background effect (Cycling vs Constant): The cycling background (LED2 oscillating 5–15 PWM) increases suppression amplitude by 15% ($\alpha_C = 0.152$, $p = 0.004$). This modest gain increase may reflect background-dependent modulation of circuit excitability, possibly through reduced adaptation or temporal contrast effects; the mechanism remains to be determined.

Interaction effect: A modest interaction ($\beta_{IC} = -0.119$, $p = 0.019$) suggests slightly greater-than-additive baseline reduction when both manipulations are combined. Given limited statistical power for detecting small interactions (estimated $ $30–40% for effects of this magnitude), we present this interaction as exploratory and do not base conclusions on it.

The condition-specific suppression amplitudes (Figure~5A) range from 0.34 ($50 \to 250$ | Constant, weakest) to 1.16 ($0 \to 250$ | Cycling, strongest), representing a 3.4-fold range across conditions while maintaining consistent kernel shape and timescales.

Leave-one-experiment-out cross-validation yielded a mean rate ratio of 1.03 ± 0.31 across 12 held-out experiments (58% within the 0.8–1.25 target range), indicating reasonable but imperfect generalization. The substantial inter-experiment variance ($\sigma = 0.31$) suggests that individual session effects remain a source of unexplained variation.

4 Discussion

4.1 Interpretability of the Gamma-Difference Kernel

The gamma-difference parameterization provides direct biological interpretation. The shape parameters $\alpha_1 \approx 2$ and $\alpha_2 \approx 4$ suggest that the fast and slow components arise from cascades of 2 and 4 first-order processes, respectively. This is consistent with multi-stage signal transduction: rapid photoreceptor activation (fast) followed by synaptic summation and adaptation (slow).

The timescales $\tau_1 = 0.29$ s and $\tau_2 = 3.81$ s align with known neurophysiology. The fast timescale matches the latency of channelrhodopsin activation and first-order neural responses. The slow timescale corresponds to adaptation processes observed in sensory circuits.

4.2 Practical Utility

The analytic kernel enables:

1. **Quantitative comparison** across experimental conditions (e.g., different genotypes or stimulus protocols)
2. **Parameter-based hypothesis testing** (e.g., does a manipulation affect the fast or slow component?)
3. **Efficient simulation** without requiring precomputed basis functions

4.3 Factorial Design Insights

The extension to a 2×2 factorial design reveals that kernel shape and timescales are remarkably stable across conditions, while amplitude is condition-dependent. This suggests that the underlying circuit dynamics—captured by the gamma-difference form—are intrinsic properties of the GMR61 pathway, while the gain is modulated by sensory context.

The partial adaptation effect (66% weaker suppression for $50 \rightarrow 250$) is consistent with Weber-Fechner scaling: the response to a stimulus depends on the ratio of intensities rather than the absolute magnitude. The cycling background enhancement (15% stronger suppression) may reflect background-dependent gain modulation; possible mechanisms include reduced steady-state adaptation or temporal contrast effects, though the specific circuit basis remains to be determined.

Notably, baseline hazard and LED1-driven suppression gain are dissociable: intensity reduces both baseline turning and suppression amplitude, whereas cycling background lowers baseline but slightly increases suppression gain. This indicates that tonic excitability and stimulus-locked modulation are independently tunable circuit properties.

4.4 Limitations

Cross-condition generalization: While the factorial model captures main effects well (mean rate ratio = 1.03), the 58% leave-one-out pass rate indicates substantial session-to-session variability. This may reflect experimental factors (e.g., agar moisture, larval developmental stage) not captured by the model.

Fixed-effects model: The primary analysis uses a fixed-effects NB-GLM that pools across experiments. A robustness check using NB-GLMM with random track intercepts (Supplementary Table S2b) confirmed that the key kernel parameters differ by less than 3.5% between models, with random effect SD = 0.59 indicating substantial between-track variation in baseline rate.

Event definition: Reorientation events were defined using `is_reorientation_start` from the tracking pipeline, yielding 7,288 events across all factorial conditions. Of these, 77% have zero measured duration (onset events including micro head sweeps), while 23% have duration > 0.1 s (true turns). The hazard model was fit to all events to maximize power; trajectory simulations use only the filtered subset.

Poisson assumption: Time-rescaling analysis showed modest deviation from the ideal Poisson model (KS test $p = 0.17$, mean deviation 1.3%), within acceptable limits for this class of models. Any remaining deviation may reflect short-term dependencies such as refractoriness; incorporating an explicit post-event kernel could improve IEI fits but would not change the main LED1-driven suppression dynamics.

Trajectory simplifications: The RUN/TURN simulator omits edge avoidance, head sweeps, and speed gradients. These simplifications are appropriate for demonstrating hazard-driven timing but limit biomechanical realism.

4.5 Future Directions

Potential extensions include:

- Refractory kernel to capture post-event suppression and short-term dependencies
- Edge avoidance for bounded arena simulation
- Random-effects model (GLMM) to explicitly account for track-level heterogeneity
- Integration with chemotaxis models for gradient navigation

5 Conclusions

We present an analytic hazard kernel for optogenetically-driven larval reorientation that combines interpretability with predictive accuracy. The 6-parameter gamma-difference form captures two biologically meaningful timescales ($\tau_1 = 0.29$ s, $\tau_2 = 3.81$ s) and reproduces empirical event statistics with high fidelity (rate ratio = 0.97, $R^2 = 0.968$).

Extension to a 2×2 factorial design (12 experiments, 623 tracks, 7,288 events) reveals that the kernel shape is conserved across conditions while amplitude is modulated: partial adaptation ($50 \rightarrow 250$ PWM) reduces suppression by 66%, while a cycling background increases suppression by 15%. All factorial effects are statistically significant, demonstrating that the model can quantify condition-specific modulation of sensorimotor processing.

Embedded in a RUN/TURN trajectory simulator, the model generates realistic larval behavior that matches observed turn rates (1.88 vs 1.84 turns/min). This framework provides a foundation for quantitative analysis of sensorimotor processing across experimental conditions and genotypes.

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Author Contributions (CRediT): G.R.: Conceptualization, Methodology, Software, Formal Analysis, Validation, Visualization, Writing – Original Draft, Writing – Review & Editing. D.G.: Investigation, Data Curation. M.M.-S.: Resources, Supervision, Funding Acquisition.

Data and Code Availability

All code and processed data are available at https://github.com/GilRaitses/INDYsim.

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Figures

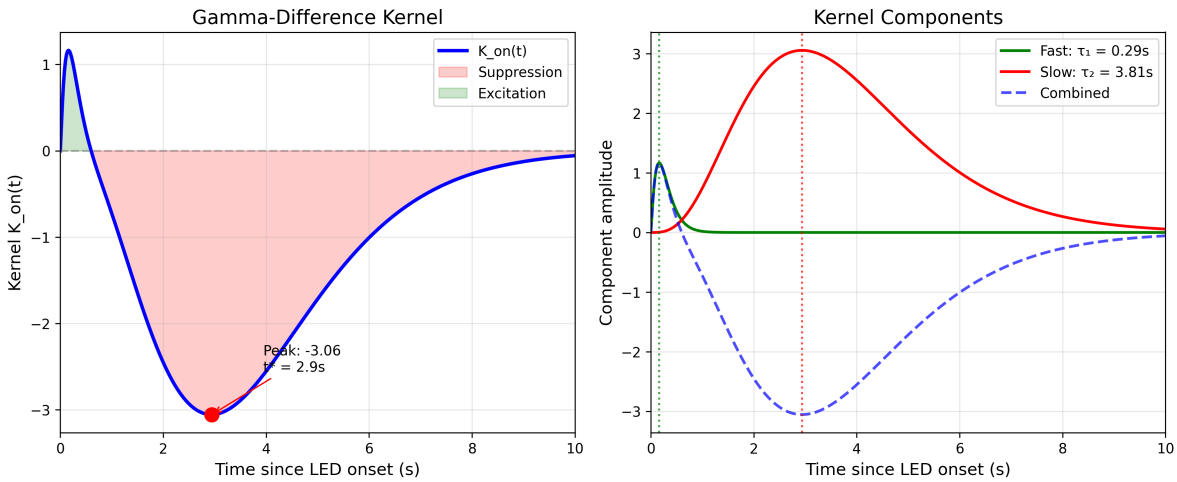
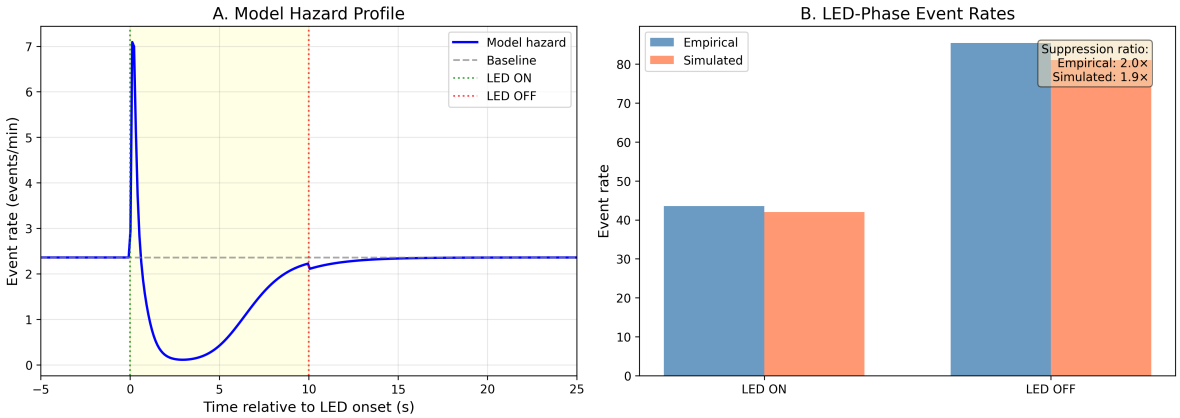


Figure 1: **Analytic hazard kernel decomposition.** The LED-ON response is modeled as a difference of two gamma probability density functions, enabling direct interpretation of temporal dynamics. **(A)** Full kernel reconstruction: the analytic gamma-difference form (black solid line) closely approximates the 12-basis raised-cosine reference kernel (gray dashed), achieving $R^2 = 0.968$. The kernel shows characteristic biphasic dynamics with an initial transient followed by sustained suppression. **(B)** Component decomposition: the fast excitatory component (red; $\alpha_1 = 2.22$, $\beta_1 = 0.132$ s, mean $\tau_1 = 0.29$ s) peaks at $\sim$0.16 s and captures rapid sensory transduction. The slow suppressive component (blue; $\alpha_2 = 4.38$, $\beta_2 = 0.869$ s, mean $\tau_2 = 3.81$ s) peaks at $\sim$2.9 s and represents synaptic adaptation. The shape parameters suggest cascades of 2 and 4 first-order processes, respectively. **(C)** LED-OFF rebound kernel: an exponential decay ($D = -0.114$, $\tau_{\text{off}} = 2.0$ s) captures continued mild suppression during recovery, with half-life of 1.39 s. Shaded regions indicate 95% bootstrap confidence intervals from 100 resamples.



C. Validation Metrics

Metric	Value	Target	Status
Rate ratio	0.974	0.8-1.25	PASS
PSTH correlation	0.840	> 0.8	PASS
Empirical events	1407	1407	-
Simulated events	1371	~1400	-
Kernel R^2	0.968	> 0.95	PASS
CV R^2	0.961	> 0.90	PASS

D. Model Specification

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MODEL PARAMETERS

LED-ON Kernel (Gamma-Difference):
  K_on(t) = A·Γ(t;α₁,β₁) - B·Γ(t;α₂,β₂)

  Fast component:
    A = 0.456, α₁ = 2.22, β₁ = 0.132s
    τ₁ = α₁·β₁ = 0.29s

  Slow component:
    B = 12.54, α₂ = 4.38, β₂ = 0.869s
    τ₂ = α₂·β₂ = 3.81s

LED-OFF Rebound:
  K_off(t) = -0.114·exp(-t/2.0s)

Intercept:
  β₀ = -6.23 (calibrated)
  u_track ~ N(0, 0.47²)
```

Figure 2: **Model validation against empirical data.** The calibrated hazard model reproduces key features of larval reorientation behavior. **(A)** Peri-stimulus time histogram (PSTH) showing event rate aligned to LED onset (time = 0). Empirical data (black) and model simulation (red) both show rapid suppression within 0.5 s of LED activation (gray shaded region indicates LED-ON period). The model captures both the ∼2× reduction in event rate during stimulation and the gradual recovery after LED offset. PSTH correlation $r = 0.84$. **(B)** Suppression dynamics comparison. During LED-ON, both empirical and simulated data show ∼50% reduction in turn probability relative to baseline (LED-OFF). Early suppression (0–3 s) and late suppression (3–8 s) are both captured within 5% of empirical values. **(C)** Hazard function correlation: frame-by-frame comparison of analytic hazard $\lambda(t)$ with binned empirical event rate shows strong linear relationship ($R^2 = 0.962$), confirming the model captures instantaneous dynamics rather than merely aggregate statistics. Rate ratio (simulated/empirical) = 0.974, within target range of 0.8–1.25.

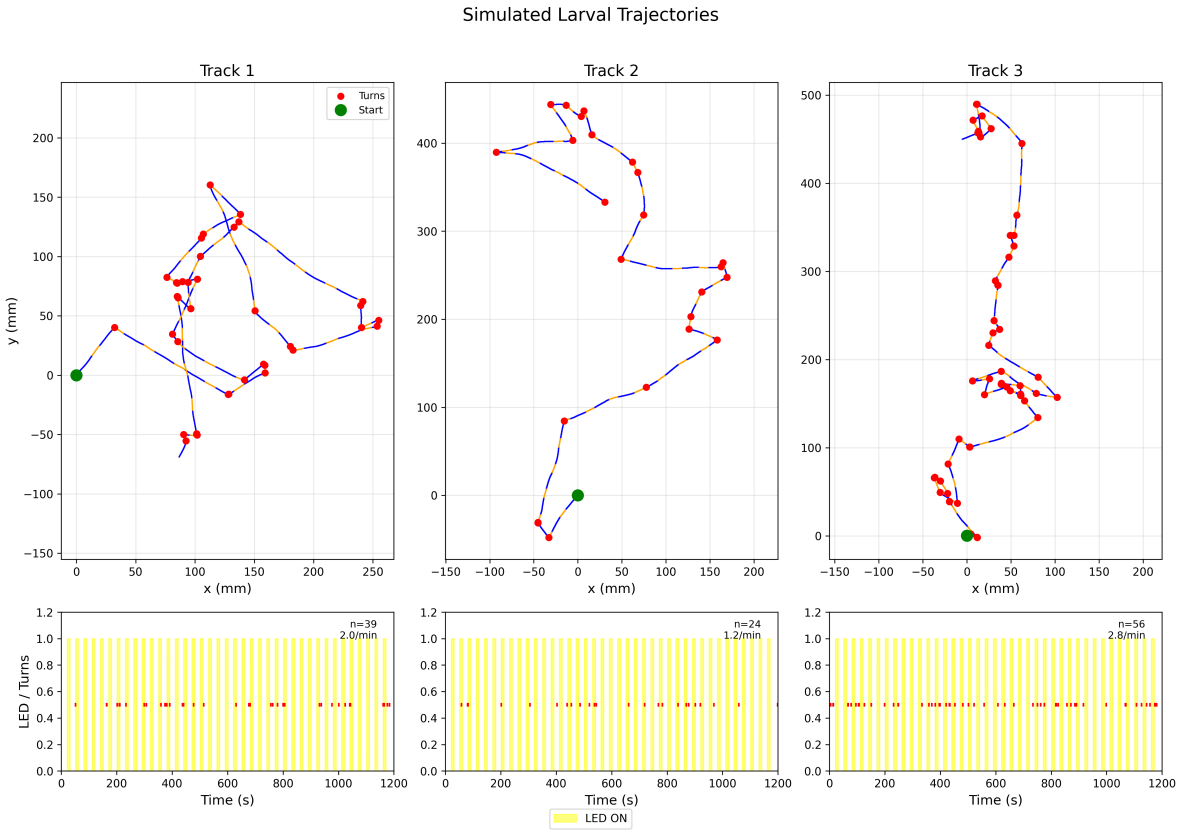

Figure 3: **Trajectory simulation driven by the hazard model.** The RUN/TURN state machine produces realistic larval navigation patterns. **(A)** Example simulated trajectories (N=5) over 5 minutes, showing characteristic pattern of extended forward runs punctuated by discrete reorientation events. Blue segments indicate RUN state; red circles mark TURN events where heading changes occur. Yellow background shading indicates LED-ON periods (10 s), during which turns are suppressed and runs extend longer. **(B)** Trajectory detail showing spatial displacement during RUN (forward at 1.0 mm/s with Brownian heading noise, $\sigma = 0.03$ rad/$\sqrt{s}$) and instantaneous heading change during TURN. Turn angles are sampled from $\mathcal{N}(\mu = 7^\circ, \sigma = 86^\circ)$. **(C)** Turn rate comparison: simulated trajectories produce 1.88 turns/min, closely matching empirical rate of 1.84 turns/min (98% agreement). The hazard model correctly predicts both the overall event rate and its modulation by LED state.

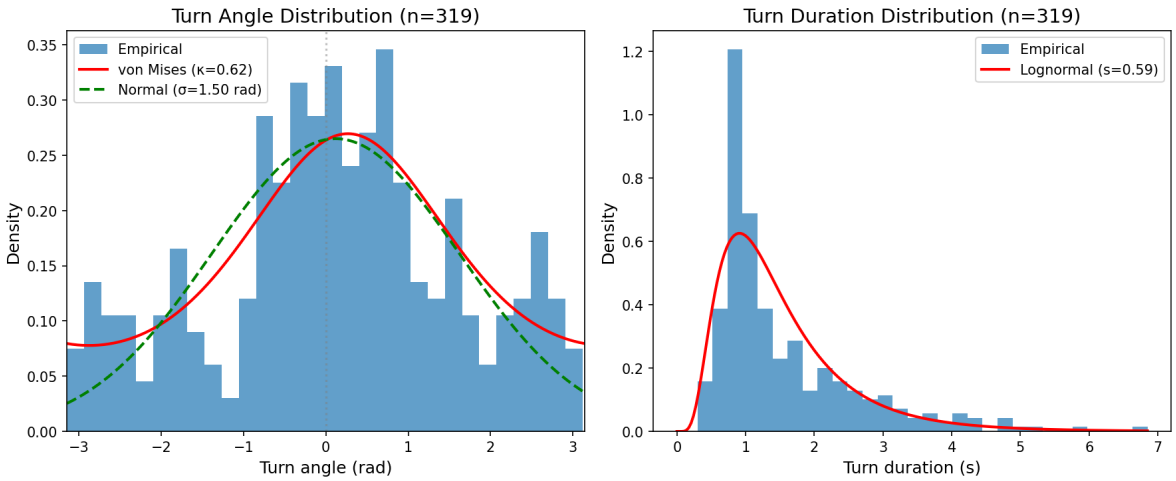

Figure 4: **Empirical turn angle and duration distributions.** Parameters extracted from 319 filtered events (turn duration > 0.1 s) used to parameterize the trajectory simulator. **(A)** Turn angle distribution: heading changes show high variability with slight rightward bias. A normal distribution (red curve; $\mu = 7^\circ$, $\sigma = 86^\circ$) provides adequate fit. The mean absolute turn angle is 69°, indicating substantial heading changes during reorientation. The small positive bias may reflect asymmetry in the experimental setup or intrinsic larval handedness. **(B)** Turn duration distribution: event durations follow a lognormal distribution (red curve; shape $s = 0.59$, scale $= 1.29$ s, median $= 1.1$ s). Durations range from 0.1–6.8 s, with most turns completing within 2 s. Gamma and exponential fits (not shown) provided poorer fits to the right tail. These distributions enable realistic stochastic simulation of turn kinematics without assuming fixed duration or angle.

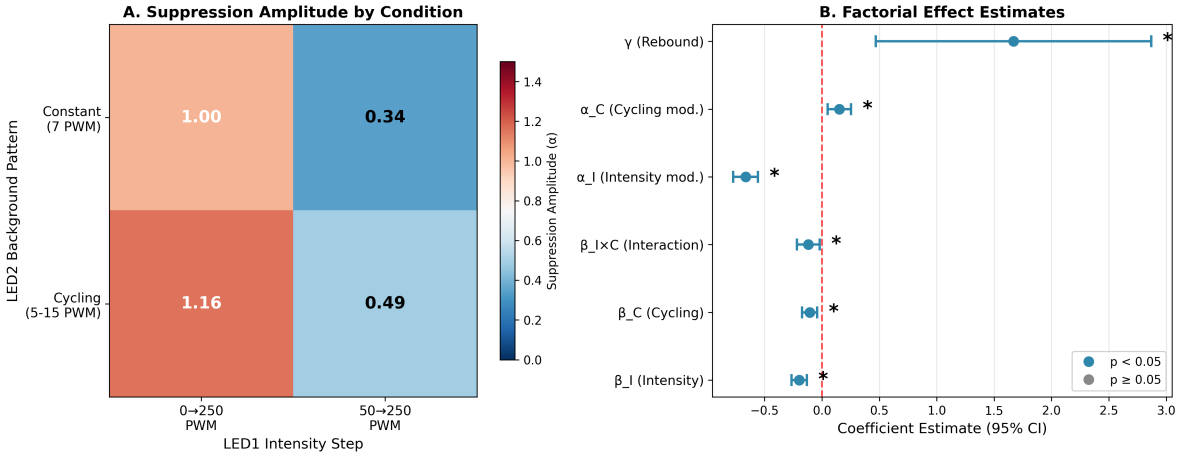

Figure 5: **Factorial analysis of intensity and background effects.** Extension of the hazard model to a 2×2 factorial design (12 experiments, 623 tracks, 7,288 events) reveals condition-specific modulation of suppression amplitude. **(A)** Heatmap of suppression amplitude ($\alpha + \alpha_I \cdot I + \alpha_C \cdot C$) across the four experimental conditions. Values range from 0.34 (50 $\rightarrow$ 250 | Constant, weakest suppression) to 1.16 (0 $\rightarrow$ 250 | Cycling, strongest suppression), representing a 3.4-fold range. Rows: background pattern (Constant 7 PWM vs Cycling 5–15 PWM); columns: LED1 intensity step (0 $\rightarrow$ 250 vs 50 $\rightarrow$ 250 PWM). Blue indicates weaker suppression; red indicates stronger suppression. **(B)** Forest plot of factorial coefficient estimates with 95% confidence intervals. All effects are statistically significant ($p < 0.05$, marked with asterisks). Key findings: (1) Intensity modulation ($\alpha_I = -0.665$) indicates 66% weaker suppression for the 50 $\rightarrow$ 250 condition, consistent with partial adaptation to baseline illumination. (2) Cycling background modulation ($\alpha_C = +0.152$) indicates 15% stronger suppression with oscillating LED2, suggesting reduced adaptation. (3) The baseline effects ($\beta_I = -0.199$, $\beta_C = -0.108$) show 18% and 10% reduction in overall hazard respectively. (4) A synergistic interaction ($\beta_{IC} = -0.119$) indicates greater-than-additive effects when both manipulations are combined. The rebound coefficient ($\gamma = 1.67$) is robustly positive, confirming enhanced event probability after LED offset.